Jones matrix of polarization mode dispersion

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We describe how to calculate the Jones matrix transfer function of a fiber if its principal states of polarization and its differential group delay as functions of frequency are known. Using two counterexamples related to second-order polarization mode dispersion (PMD), we also show that a previous method used for the same purpose induces overestimation of second-order PMD effects by a factor of 2. Our new method is used to solve the problem for both counterexamples. © 1999 Optical Society of America

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To the first order, polarization mode dispersion (PMD) is a group-velocity difference between two orthogonal states of polarization.[1] These orthogonal states are called the principal states of polarization (PSP's), and the difference in arrival time between both axes is called the differential group delay (DGD). PMD is a statistical process in optical fibers. Owing to variation of the local birefringence, the DGD and the PSP's vary with time. The mean value of the DGD is also called PMD. It has been shown that, as long as the PMD is lower than a few percent (for instance, 10%) of the bit duration of a nonreturn-to-zero digital signal, PMD conditions (i.e., the DGD and the PSP's) can be considered constant over the spectrum of the signal. Because of the increase of the bit rates and the high PMD values of installed fibers, the variations of the PSP's and the DGD over the spectrum cannot be neglected.[2] These variations are known as higher-order PMD. Therefore it is important to be able to simulate the distortion of a signal if the PMD conditions of the fiber versus frequency are known. To do so we need to compute the transfer function of the fiber, where its DGD and PSP's versus frequency are known. This transfer function can be expressed as a 2×2 complex matrix $\mathbf{M}(\omega)$, called a Jones matrix, that transforms the two complex components of the electrical field from the input of the fiber to the output in the Fourier domain. Computing the Jones matrix that corresponds to PMD is a problem that was easily solved for first-order PMD, and a method has been suggested for higher-order PMD terms.[3] First we use two counterexamples to show that this method was wrong. The first is a linear variation of the DGD versus frequency and constant PSP's, and the second is a constant rotation of the PSP's versus frequency and a constant DGD. These examples correspond to both effects of second-order PMD.[3] We then explain that, even though the mistake in the previous method induces magnification by a factor of 2 of second-order PMD effects, the main conclusions of the studies in which the method given in Ref. 3 was used are still valid. We then give the equations that have to be used to find the exact Jones matrix when the DGD and the PSP's versus frequency are known. These equations are solved for both counterexamples.

To the first order, PMD is a DGD between two orthogonal PSP's and can be expressed with the Jones matrix $\mathbf{M}(\omega) = \mathbf{R}\mathbf{P}(\omega)\mathbf{R}^{-1}$, where $\mathbf{R}$ is a rotation matrix that is used to turn the optical field into the frame of the PSP's and $\mathbf{P}(\omega)$ is a diagonal matrix that corresponds to the DGD:

$$\mathbf{R} = \begin{bmatrix} \cos\theta\cos\epsilon - i\sin\theta\sin\epsilon & -\sin\theta\cos\epsilon + i\cos\theta\sin\epsilon \\ \sin\theta\cos\epsilon + i\cos\theta\sin\epsilon & \cos\theta\cos\epsilon + i\sin\theta\sin\epsilon \end{bmatrix}, \qquad \mathbf{P}(\omega) = \begin{bmatrix} \exp\left(\frac{i}{2}\,\omega\mathrm{dgd}\right) & 0 \\ 0 & \exp\left(-\frac{i}{2}\,\omega\mathrm{dgd}\right) \end{bmatrix},$$

where ω is the angular frequency, θ and ϵ are the azimuth and the ellipticity of one of the PSP's, and dgd is the constant DGD. To take into account higher-order PMD, it was suggested in Ref. 3 that one could replace dgd with $\mathrm{dgd}(\omega)$ and θ and ϵ with $\theta(\omega)$ and $\epsilon(\omega)$ in $\mathbf{R}$ and $\mathbf{P}$. This replacement seems natural. However, it is wrong because there is confusion between the phase and its derivative. To make this confusion obvious we consider two counterexamples. In case (a) we consider linear variation of the DGD $[\mathrm{dgd}(\omega) = \mathrm{dgd}_0 + \mathrm{dgd}_1\omega]$ and constant PSP's ($\theta = \theta_0$ and $\epsilon = \epsilon_0$). In case (b) we consider $\mathrm{dgd} = \mathrm{dgd}_0$, $\theta = k\omega$, and $\epsilon = 0$, where k is the supposed constant rotation rate of the PSP's. In an actual fiber it is almost impossible to generate these particular counterexamples. Nevertheless, as a first

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approximation, when the DGD varies, its variation can be approximated as a linear variation [case (a)]. In the same way, when the PSP's vary, their variation can be approximated as a constant rotation over a great circle of the Poincaré sphere, which can be changed into the equator by a change of frame [case (b)]. We use uppercase letters to express the actual values of the parameters and lowercase letters for the parameters that are used to fill the Jones matrices **R** and **P**. The actual group delay of $\mathbf{M}(\omega)$ in case (a) is equal to the frequency derivative of the phase difference:

$$\mathrm{DGD}_a(\omega) = \frac{\partial}{\partial\omega}[\omega\,\mathrm{dgd}(\omega)] = \mathrm{dgd}_0 + 2\,\mathrm{dgd}_1\omega\,.$$

Hence the actual derivative of the DGD is equal to $2\,\mathrm{dgd}_1$ and not dgd_1. The influence of the derivative of the DGD is therefore overestimated by a factor of 2 in Ref. 3.

The actual DGD and the PSP's are equal to the difference of the modulus of the eigenvalues of $\mathbf{M}^{-1}\partial\mathbf{M}/\partial\omega$ and to its eigenvectors,[1] respectively. Using this method for case (b), we found that

$$\mathrm{DGD}_b(\omega) = \left[\mathrm{dgd}_0 + 16k^2\sin^2\!\left(\frac{\mathrm{dgd}_0\,\omega}{2}\right)\right]^{1/2}.$$

Figure 1 shows the actual DGD, $\mathrm{DGD}_b(\omega)$, versus frequency for realistic values of dgd_0 and k of a fiber that exhibits a PMD of 30 ps.[4]

The actual DGD is not constant, although $\mathrm{dgd} = \mathrm{dgd}_0$ is constant. However, $\mathrm{DGD}_b(0) = \mathrm{dgd}_0$. Furthermore, computing the eigenvectors of $\mathbf{M}^{-1}\partial\mathbf{M}/\partial\omega$, developing the PSP's into a series around $\omega = 0$, and using the relation between the azimuth and the ellipticity and the Jones representation of a SOP,[5] we obtained

$$\Theta_b(\omega) = 2k\omega + o(\omega^2)\,,$$

$$E_b(\omega) = \frac{1}{2}\,k\,\mathrm{dgd}_0\,\omega^2 + o(\omega^2)\,.$$

Therefore the actual PSP's do not remain on the equator of the Poincaré sphere. These results were confirmed by our numerical simulations: Figure 2 represents one of the PSP's versus frequency in the same condition as in Fig. 1. Furthermore, the actual rotation rate $K(\omega)$ of the PSP's varies with frequency and is equal to $2k$ near 0. Thus the influence of second-order PMD was also overestimated by a factor of 2 in case (b).

To our knowledge all the papers in which the Jones matrix $\mathbf{M}(\omega) = \mathbf{R}(\omega)\mathbf{P}(\omega)\mathbf{R}(\omega)^{-1}$ was considered in evaluations of the effect of second-order PMD on system performance have overestimated second-order PMD effects. Furthermore, what was considered second-order PMD was not pure second-order PMD only. Nevertheless, the main conclusions of these papers (chromatic dispersion has a strong effect in the presence of higher-order PMD, second-order PMD induces overshoots of the edges of nonreturn-to-zero

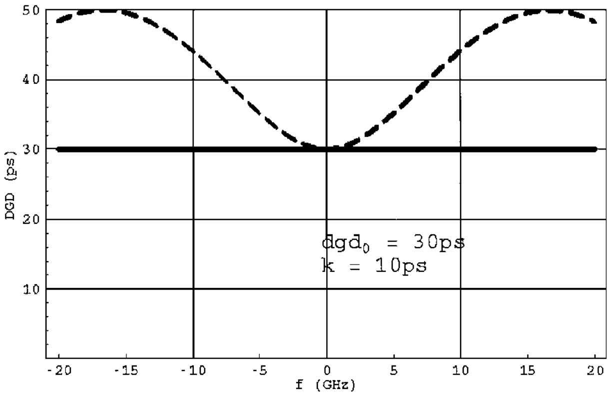

Fig. 1. Assumed (solid line) and actual [$\mathrm{DGD}_b(\omega)$, dashed curve] DGD as a function of frequency for counterexample (b).

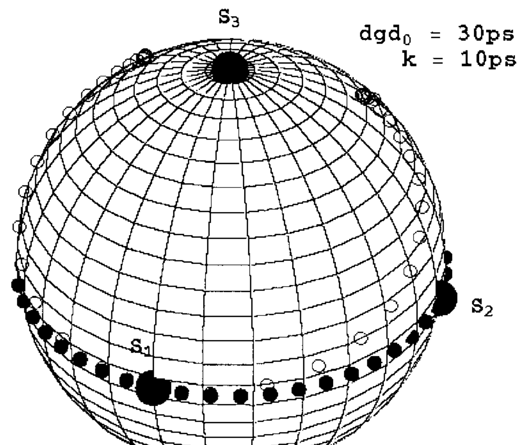

Fig. 2. Assumed (filled circles) and actual (open circles) PSP's over the Poincaré sphere as a function of frequency for counterexample (b). These values are typical for a fiber that exhibits 30-ps PMD. The circles are plotted every 1 GHz. The actual rotation rate of the PSP's over the Poincaré sphere is equal not to $2k$ but to $4k$, yielding an overestimation by a factor of 2 of second-order PMD effects.

pulses, etc.) remain valid.[4,6] For instance, the first-order derivative of the DGD has a moderate effect compared with that of the rotation of the PSP's and higher-order PMD effects are enhanced by chirp or chromatic dispersion. Below we show how to calculate the exact Jones matrix $\mathbf{M}(\omega)$.

If one knows the DGD and the PSP's of a fiber at any frequency, as can be found with a polarization analyzer, the aim is to compute the transfer function of the fiber as a Jones matrix $\mathbf{M}(\omega)$. Note that many solutions exist. Consider a rotation matrix **R** that is independent of ω; then $\mathbf{M}(\omega)$ and $\mathbf{RM}(\omega)$ have the same DGD and PSP's because $(\mathbf{RM})^{-1}\partial(\mathbf{RM})/\partial\omega = \mathbf{M}^{-1}\partial\mathbf{M}/\partial\omega$.

In case (a), a simple solution is to replace the phase in $\mathbf{P}(\omega)$, $\omega\,\mathrm{dgd}/2$, with the integral of $\mathrm{DGD}(\omega)/2$. For the general case we inverted the theory explained in Ref. 1 and found that

$$\frac{\partial\mathbf{M}(\omega)}{\partial\omega} = \mathbf{M}(\omega)\mathbf{A}(\omega)\,,$$

where

$$\mathrm{M}(\omega) = \begin{bmatrix} m_1(\omega) & m_2(\omega) \\ -m_2{}^*(\omega) & m_1{}^*(\omega) \end{bmatrix},$$

$$\mathrm{A}(\omega) = \begin{bmatrix} a_1(\omega) & a_2(\omega) \\ -a_2{}^*(\omega) & a_1{}^*(\omega) \end{bmatrix}$$

$$= \mathbf{R}(\omega) \begin{bmatrix} \frac{i}{2}\,\mathrm{DGD}(\omega) & 0 \\ 0 & -\frac{i}{2}\,\mathrm{DGD}(\omega) \end{bmatrix} \mathbf{R}^{-1}(\omega).$$

$\mathbf{R}(\omega)$ has the same form as matrix $\mathbf{R}$ above and corresponds to the actual PSP's at frequency ω. Developing the matrix differential equation, we obtain

$$\begin{bmatrix} m_1{}'(\omega) \\ m_2{}'(\omega) \end{bmatrix} = \begin{bmatrix} a_1(\omega) & -a_2{}^*(\omega) \\ a_1(\omega) & a_1{}^*(\omega) \end{bmatrix} \begin{bmatrix} m_1(\omega) \\ m_2(\omega) \end{bmatrix}, \quad (1)$$

which is a system of two coupled first-order differential equations with varying coefficients. The initial conditions are the DGD and the position of the PSP's at = 0. Since the PSP's, i.e., the eigenvectors of $\mathbf{M}^{-1}\partial\mathbf{M}/\partial\omega$, can be multiplied by any constant and remain PSP's, an infinite number of solutions exist, as shown above.

In case (b), $a_1(\omega) = i\ \mathrm{dgd}_0/2\cos(2k\omega)$ and $a_2(\omega) = i\ \mathrm{dgd}_0/2\sin(2k\omega)$. To solve Eq. (1) we set $u(\omega) = m_1(\omega) + im_2(\omega)$ and $v(\omega) = m_1(\omega) - im_2(\omega)$. Using Eq. (1), we can show that $u' = i\ \mathrm{dgd}_0/2\exp(i2k\omega)v$ and $v' = i\ \mathrm{dgd}_0/2\exp(-i2k\omega)u$. Therefore $u'' - i2ku' - \mathrm{dgd}_0{}^2/4u = 0$ and $v'' + i2kv' - \mathrm{dgd}_0{}^2/4v = 0$. This gives us u and v with two complex integration constants, C_1 and C_2, and therefore m_1 and m_2 are

$$m_1(\omega) = C_1\left[\exp(i\alpha_-\omega) + 2\frac{\alpha_-}{\mathrm{dgd}_0}\exp(-i\alpha_+\omega)\right] + C_2\left[\exp(i\alpha_+\omega) + 2\frac{\alpha_+}{\mathrm{dgd}_0}\exp(-i\alpha_-\omega)\right],$$

$$m_2(\omega) = iC_1\left[-\exp(i\alpha_-\omega) + 2\frac{\alpha_-}{\mathrm{dgd}_0}\exp(-i\alpha_+\omega)\right] + iC_2\left[-\exp(i\alpha_+\omega) + 2\frac{\alpha_+}{\mathrm{dgd}_0}\exp(-i\alpha_-\omega)\right],$$

with

$$\alpha_\pm = k \pm \frac{1}{2}(\mathrm{dgd}_0{}^2 + 4k^2)^{1/2}.$$

Setting the initial PSP's to (1, 0) and (0, 1) at $\omega = 0$ gives one solution for case (b):

$$C_1 = -\frac{1}{2}\exp\left(-i\frac{\pi}{4}\right)\frac{1}{(\mathrm{dgd}_0{}^2 + 4k^2)^{1/2}}\left(i\frac{\mathrm{dgd}_0}{2} + \alpha_+\right),$$

$$C_2 = \frac{1}{2}\exp\left(+i\frac{\pi}{4}\right)\frac{1}{(\mathrm{dgd}_0{}^2 + 4k^2)^{1/2}}\left(\frac{\mathrm{dgd}_0}{2} - i\alpha_-\right).$$

We verified that $\mathbf{M}(\omega)$ with this solution effectively has a constant DGD equal to dgd_0 and PSP's defined by $\theta = k\omega$ and $\epsilon = 0$.

Using two counterexamples, we have shown that the often-suggested use of the Jones matrix to describe PMD when the DGD and the PSP's versus frequency are known is wrong in the general case. One of the consequences of this approach is that the effect of second-order PMD on system performance has often been overestimated by a factor of 2. However, most conclusions of the paper is which this formalism was used were correct. Furthermore, we have shown a method to compute the correct matrix and given the matrices for out two counterexamples.

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